

Experiment No. 1

Title: PERFORM TENSION TEST ON MILD STEEL.

Objective: **Object:** - To conduct a tensile test on a mild steel specimen and determine the following:

- i) Ultimate strength ii) Upper yield strength iii) Lower yield strength
- iv) Percentage elongation

Equipment used: Universal testing machine

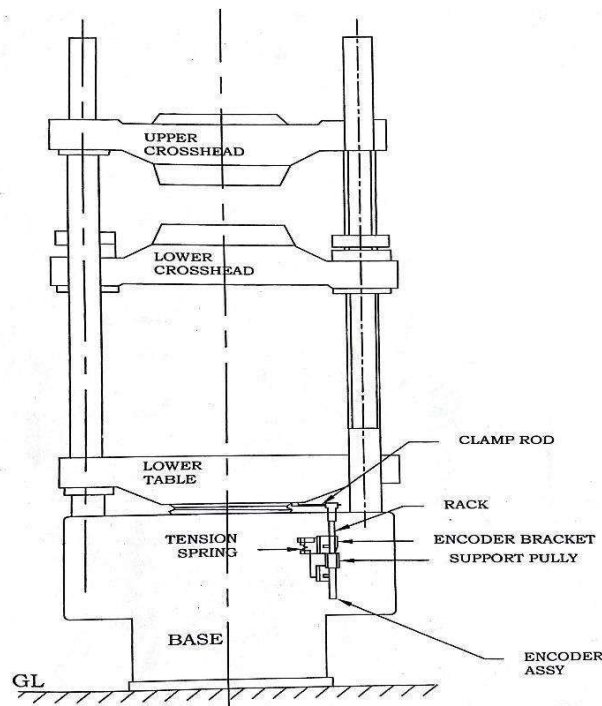


Fig. 1 Schematic of Universal Testing Machine

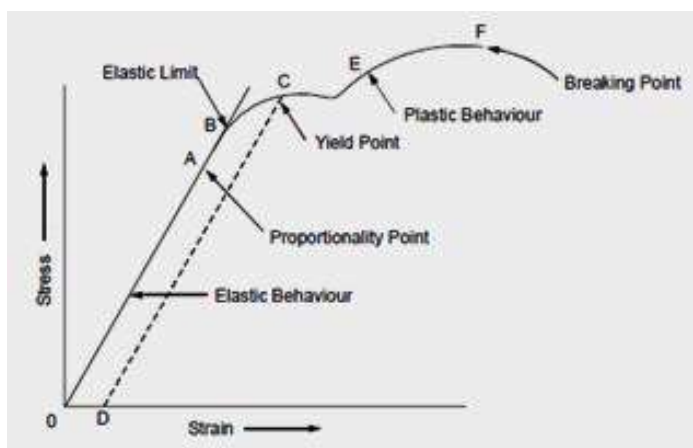
Description:

Tensile testing is one of the more basic tests to determine stress – strain relationships. It is a simple uniaxial test consists of slowly pulling a specimen in tension until it breaks. Test specimens for tensile testing are generally either circular or rectangular with larger ends to facilitate gripping the sample. During testing, the stress in the specimen can be calculated by dividing the load over the cross-sectional area and the strain calculated by taking the change in length and dividing it by the original or initial length. The stress and strain measurements and calculations discussed so far assume a fixed cross sectional of the specimen. These stress and strain values are known as engineering stress and engineering strain. Most of these engineering values are found by graphing the stress and strain values from testing. The modulus of elasticity can be calculated by finding the slope of the stress strain

curve where it remains linear and constant. Again, where the stress strain curve starts to become nonlinear, that point is known as the proportional limit. The proportional limit is also the point where yielding occurs in the material. At this point, the material no longer exhibits elastic behavior and permanent deformation occurs. This onset of inelastic behavior is defined as the yield stress or yield strength. Some materials such as the mild steel will have a well-defined yield point that can be easily identified on the stress strain curve. The highest stress or load the material is capable of will be the highest measurable stress on the graph. This is termed the ultimate strength or tensile strength of the specimen. The point at which the material actually fractures is termed the fracture stress. For ductile materials, the Ultimate stress is greater than the fracture stress, but for brittle materials, the ultimate stress is equal the fracture stress.

Procedure:

1. Measure the diameter of the specimen at least three different places and determine the mean value. Also measure its gauge length
2. Fix the specimen between the upper and middle cross head jaws of the machine
3. Close the release valve on the machine and turn on the machine
4. Now apply load on the specimen by turning slowly the load valve in anticlockwise direction until the specimen fails.
5. Note down the load and its corresponding length. The observations are taken until the specimen breaks.
6. Measure the final length and diameter of specimen at the fracture by joining the two broken halves of the specimen together
7. Now calculate the desired values and plot a graph between stress vs. strain



ig. 3 Stress-Strain curves for ductile material

Observation Table: (1)

Sl. No.	Initial diameter ' D_1 ' (mm)	Final diameter ' D_2 ' (mm)	Initial length ' L_1 ' (mm)	Final length ' L_2 ' (mm)	Initial cross section area ' A_1 ' (mm ²)	Area at fracture ' A_2 ' (mm ²)

Observation Table: (2)

Sl. No.	Load ' P ' (kN)	Displacement ' δl ' (mm)	Stress ' σ ' (kN/mm ²)	Strain ' e_t '	Modulus of elasticity ' E ' (kN/mm ²)

Calculations:

1.	Stress	=	$\frac{P}{A_1} \text{ kN/mm}^2$
2.	Strain	=	$\frac{\delta l}{L_1}$
3.	Modulus of Elasticity	=	$\frac{\sigma}{e_t} \text{ kN/mm}^2$
4.	Ultimate tensile strength	=	$\frac{\text{Ultimate load}}{A_1} \text{ kN/mm}^2$
5.	Percentage elongation	=	$\frac{L_2 - L_1}{L_1} \times 100\%$
6.	Percentage reduction in area	=	$\frac{A_1 - A_2}{A_1} \times 100\%$

Conclusion:

Experiment No. 2

Title: Determination of compressive strength of a concrete cube

Objective: To find out the compressive strength of a concrete cube by compression test

Equipment used: Universal testing machine (UTM)

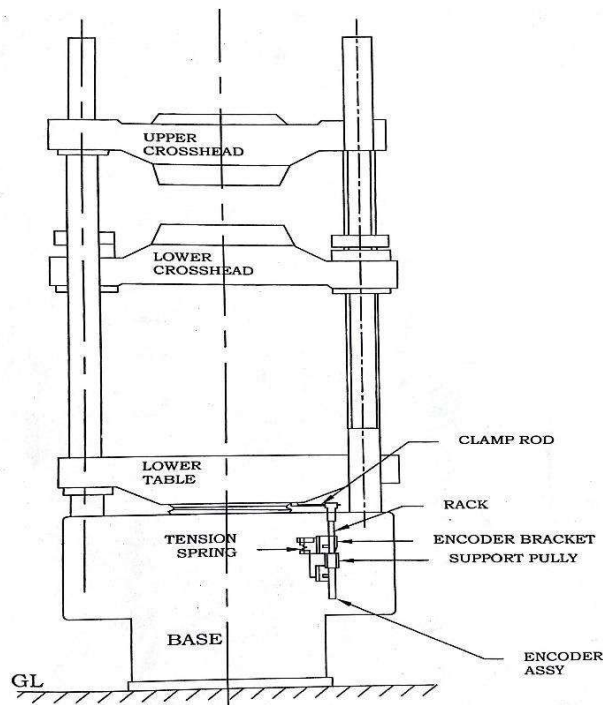


Fig. 1 Schematic diagram of Universal Testing Machine

Theory:

A compression test determines behavior of materials under crushing loads. Several machines and structure components such as columns and struts are subjected to compressive load. These components are made of high compressive strength materials. Not all the materials are strong in compression. Several materials, which are good in tension, are poor in compression. Contrary to this, many materials poor in tension but very strong in compression. That is why determination of ultimate compressive strength is essential before using a material. This strength is determined by conduct of a compression test which is just opposite in nature to tensile test. Nature of deformation and fracture is quite different from that in tensile test. In this test a compressive load is applied on the cube which tends to squeeze the specimen. The load is applied on the cube till its failure. The

ratio between the load at failure and the cross-sectional area of the cube will give the ultimate compressive strength of the cube.

Procedure:

1. Measure the dimension of the cube and calculate the cross-section area
2. Place the cube centrally on the lower table, such that the centre of the lower crosshead is vertically above the centre of specimen.
3. Close the release valve on the machine and turn on the machine
4. Now apply load on the specimen by turning slowly the load valve in anticlockwise direction until the specimen fails
5. Note down the load at which the specimen breaks from the display of the machine and open the release valve.
6. Now draw a stress-strain curve by using the values obtained during the test

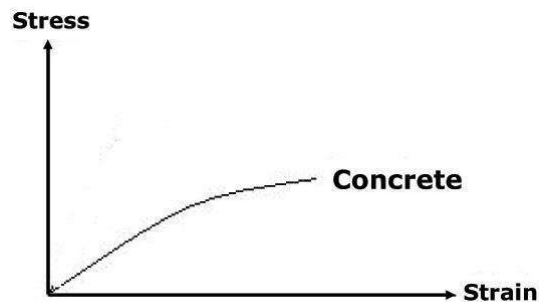


Fig. 2 Stress-Strain curve for concrete

Observation & Calculation:

- | | |
|--|-----------------------------------|
| 1. Length of the cube 'l' | = _____ mm |
| 2. Breadth of the cube 'b' | = _____ mm |
| 3. Load taken by the cube at the time of failure 'W' | = _____ N |
| 4. Cross-section area of the cube 'A' | = _____ mm ² |
| 5. Compressive strength or Ultimate strength of the cube ' $\sigma_{\max}$ ' | = $\frac{W}{A}$ N/mm ² |

Result: The compressive strength of given concrete cube is _____ N/mm²

Experiment No. 4

Title: Determination of impact strength by Charpy impact test

Objective: To determine the resistance of shock or impact strength of a given specimen by Charpy impact test

Equipments used: Impact testing machine

Theory:

The impact is a strong force or shock which occurs over a short time period when two or more bodies collide. Such a force or acceleration usually has a greater effect than a weaker force applied over a proportionally longer period. The impact test is performed to find out the energy needed to fracture a material and its toughness. The toughness of a material is its ability to absorb energy in plastic range especially when subjected to dynamic loading. The impact test can also be used to determine the ductility of a material. If the fracture surface is a flat plane, the material is brittle, and if the material breaks with jagged edges or shear lips, then the material is ductile. Usually a material does not break in just one way or the other, and thus comparing the jagged to flat surface areas of the fracture will give an estimate of the percentage of ductile and brittle fracture.

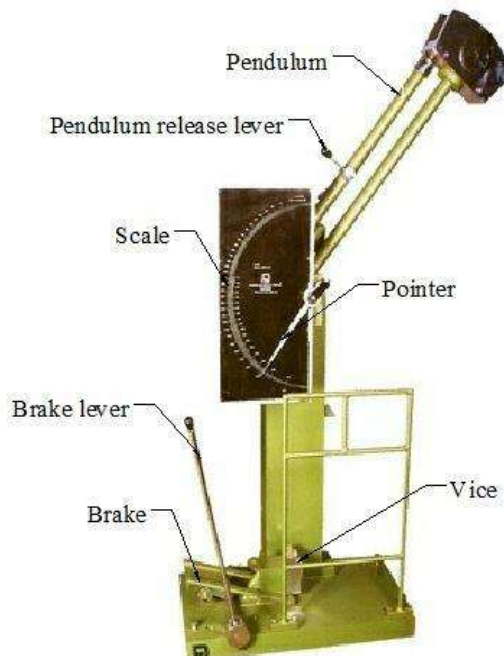


Fig. 1 Impact testing machine

In Charpy impact test, the specimen is placed as 'simply supported beam' between two supports with a notch equidistant between them. It is struck by the pendulum released from an angle of 135° , on the side opposite to the notch. In this test two types of specimen are in common use - the keyhole notch specimen and the V notch specimen. In this experiment the V-notch specimen is used and it is a bar of $10\text{ mm} \times 10\text{ mm}$ cross section which has a 45° V-notch, 2 mm deep with a 0.25 mm root radius as shown in Fig 2. The specimen is supported as a beam in horizontal position, with span 40 mm, and is struck behind the notch by the impact of a heavy swinging pendulum.

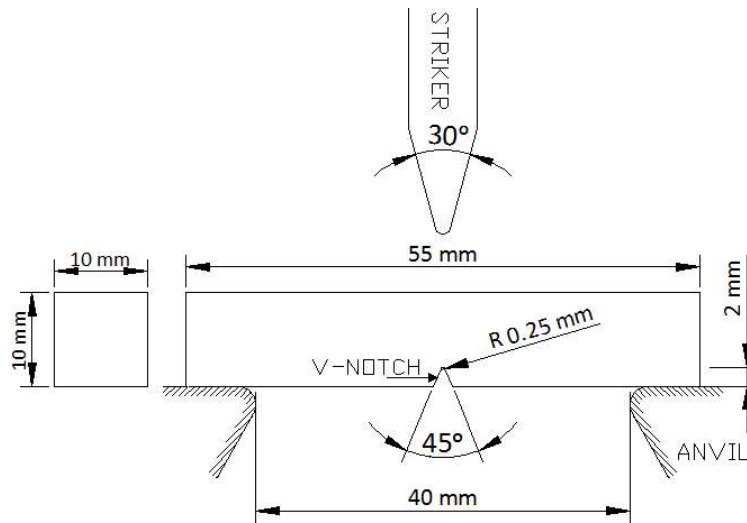


Fig. 2 Dimensions of specimen for Charpy impact test

The Notch Impact strength (I) is calculated as per the following formula

$$I = \frac{K}{A},$$

where, I = Impact strength in J/m²

K= Impact energy absorbed by the specimen in J

A= Area of cross section of specimen below the notch before test in m²

Procedure:

1. The scale of the machine is calibrated by releasing the hammer from the clamp without placing any specimen on the anvil.
2. The hammer is brought back to its initial energy level and clamped.
3. The specimen is fixed as a simply supported beam in the anvil.
4. The scale is adjusted to read zero.

5. The hammer is released from the clamp.
6. The energy absorbed by the specimen during fracture is noted from the scale.

Observation and Calculation:

Specimen Material	= _____
Impact energy absorbed (reading on the scale) 'K'	= _____ J
Area of cross section of specimen below the notch 'A'	= _____ m ²
The notch impact strength 'I'	= _____ J/m ²

Conclusion:

From this experiment we found,

Impact Strength (I) for the specimen by Charpy impact test is _____ J/m²

Experiment No. 5

Title: Determination of impact strength by Izod Impact Test

Objective: To determine the resistance of shock or impact strength of a given specimen by Izod impact test

Equipments used: Impact testing machine

Theory:

The impact is a strong force or shock which occurs over a short time period when two or more bodies collide. Such a force or acceleration usually has a greater effect than a weaker force applied over a proportionally longer period. The impact test is performed to find out the energy needed to fracture a material and its *toughness*. The *toughness* of a material is its ability to absorb energy in plastic range especially when subjected to dynamic loading. The impact test can also be used to determine the ductility of a material. If the fracture surface is a flat plane, the material is brittle, and if the material breaks with jagged edges or shear lips, then the material is ductile. Usually a material does not break in just one way or the other, and thus comparing the jagged to flat surface areas of the fracture will give an estimate of the percentage of ductile and brittle fracture.

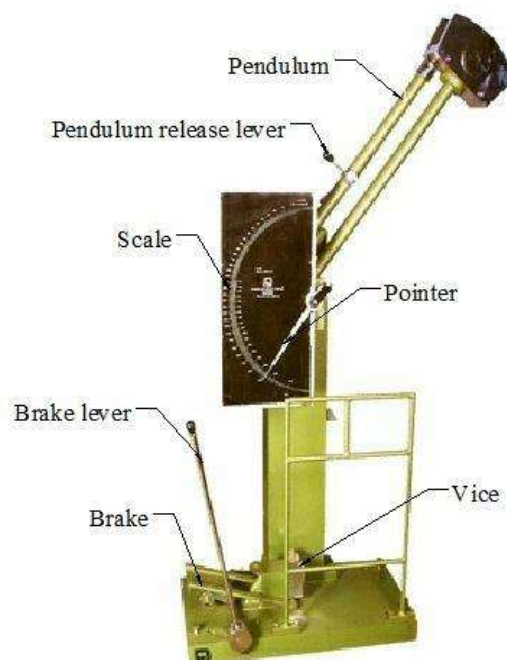


Fig 1 Impact testing machine

In Izod impact test, the specimen is held in a vice as a cantilever beam, with the center line of the notch at the same level as the top surface of the clamping device. It is struck by the pendulum released from an angle of 90° on the side of the notch 22 mm above the edge of the clamp. In this test two types of specimen are in common use - the keyhole notch specimen and the V notch specimen. In this experiment the V-notch specimen is used and it is a bar of $10\text{ mm} \times 10\text{ mm}$ cross section which has a 45° V-notch, 2 mm deep with a 0.25 mm root radius as shown in Fig 2.

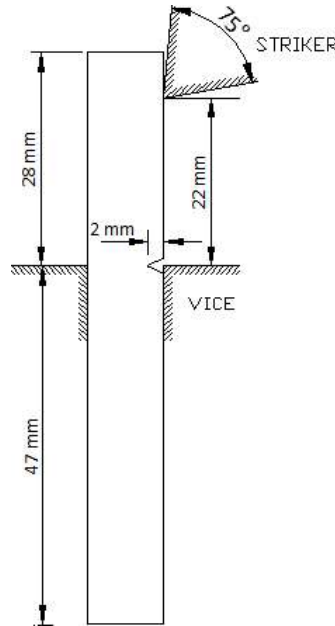


Fig. 2 Dimensions of specimen for Izod impact test

The Notch Impact strength (I) is calculated as per the following formula

$$I = \frac{K}{A}$$

where,

I = Impact strength in J/m²

K= Impact energy absorbed by the specimen in J

A = Area of cross section of specimen below the notch in m²

Procedure:

1. The scale of the machine is calibrated by releasing the hammer from the clamp without placing any specimen on the anvil.
2. The hammer is brought back to its initial energy level and clamped.
3. The specimen is fixed as a simply supported beam in the anvil.

4. The scale is adjusted to read zero.
5. The hammer is released from the clamp.
6. The energy absorbed by the specimen during fracture is noted from the scale.

Observation and calculation:

Specimen material	= _____
Impact energy absorbed (reading on the scale) 'K'	= _____ J
Area of cross section of specimen below the notch 'A'	= _____ m ²
The notch impact strength 'I'	= _____ J/m ²

Conclusion:

From this experiment we found,

Impact Strength (I) for the specimen by Izod test is _____ J/m²

Experiment No. 6

Title: Determination of hardness by Brinell hardness test

Objective: To determine the hardness of a given specimen by Brinell hardness test

Equipments used: Hardness Testing Machine

Theory:

Hardness is defined as the resistance against plastic deformation or ability to resist abrasion, scratching and indentation. In Brinell hardness test a hard steel ball of diameter '**D**' is pressed into the surface of the specimen by a load '**P**', which gives an indentation mark of the steel ball on the surface as indicated on Fig 2. The diameter of the indentation '**d**' is measured with the help of a microscope and the Brinell hardness number (BHN) is found out by the following equation,

$$\begin{aligned} \text{BHN} &= \frac{\text{Applied load}}{\text{Area of Intendation}} \\ &= \frac{2P}{\pi D(D - \sqrt{D^2 - d^2})} \end{aligned}$$



Fig. 1 Hardness testing machine

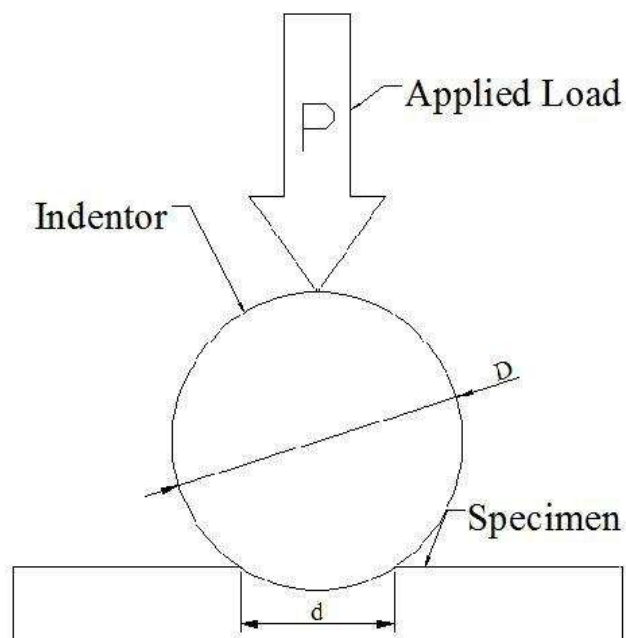


Fig. 2 Brinell hardness test

Procedure:

1. Clean the surface of the material so that it is free from dirt.
2. Put the steel ball indenter (of suitable diameter) into position.
3. Keep the lever at position 'A'.
4. Place specimen securely on anvil and set the load.
5. Turn the hand wheel clockwise, so that specimen will push the indenter and show a reading on dial gauge, as small pointer reaches at '3' (red spot) and long pointer reaches at '0' or 'Set' position.
6. Turn the lever from position 'A' to 'B' slowly so that the total load is brought into action without any jerks. The long pointer of dial gauge reaches a steady position when indentation is complete.
7. The load is allowed to act for 10 – 15 seconds.
8. Take back the lever to position 'A' slowly.
9. The specimen is taken out by turning the hand wheel anticlockwise.
10. The diameter of the impression is measured using microscope.

Observation Table:

Sl. No.	Diameter of steel ball 'D' (mm)	Load applied 'P' (kgf)	Diameter of indentation 'd' (mm)	BHN

Calculation:

$$\begin{aligned} \text{BHN} &= \frac{\text{Load applied (P)}}{\text{Spherical area of impression}} \\ &= \frac{2P}{\pi D (D - \sqrt{D^2 - d^2})} \end{aligned}$$

Conclusion:

From the experiment we found,

The Brinell hardness number (BHN) of the specimen is _____ kgf/mm²

Experiment No. 7

Title: Determination of modulus of rigidity, torsional stiffness and ultimate shear strength by Torsion Test

Objective: To determine the Modulus of Rigidity, Torsional Stiffness and Ultimate Shear Strength of a given specimen

Equipments Used: Torsion Testing Machine

Theory:

In workshops and factories, a turning force is always applied to transmit energy by rotation. This turning force is applied to a suitable point at some distance from the axis of the shaft. The product of this turning force, and the distance between the point of application of the force and the axis of the shaft is known as torque, which is also known as turning moment or twisting moment. So whenever a twisting force or torque is applied on a shaft, then the shaft is said to be subjected to torsion. Due to this torque, every cross section of the shaft is subjected to some shear stress.

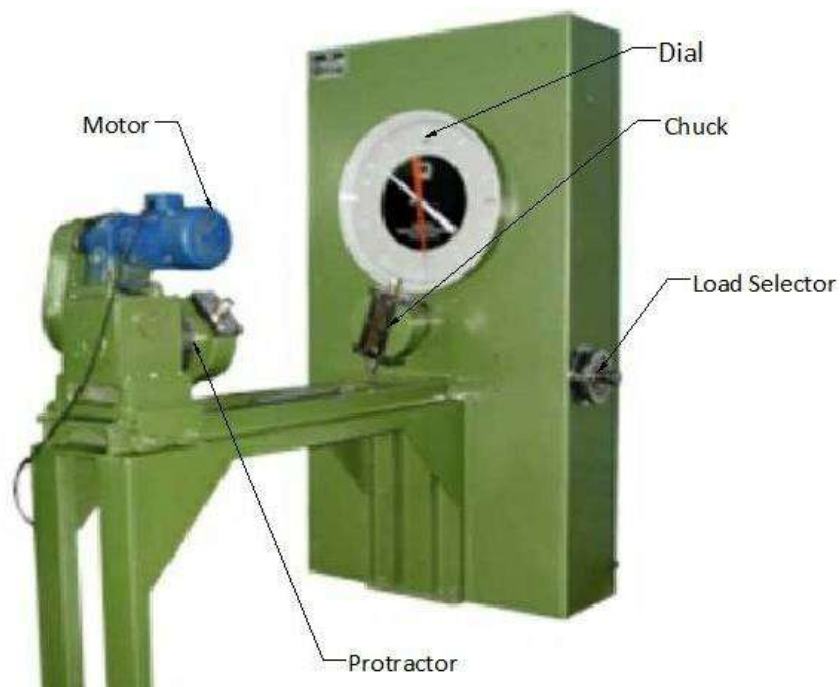


Fig 1 Torsion testing machine

In Torsion test, if a solid shaft of diameter 'D' is subjected to torque 'T' whose angle of twist is 'θ' (as shown in Fig. 2), then modulus of rigidity, shear stress and torsional stiffness of the shaft is calculated by the torsion equation as follows:

$$\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{L}$$

where,

J = Polar moment of inertia of the cross-section of the shaft

$$= \frac{\pi D^4}{32} \text{ mm}^4$$

τ = Shear stress on the shaft

L = Length of the shaft (gauge length)

G = Modulus of rigidity of the shaft

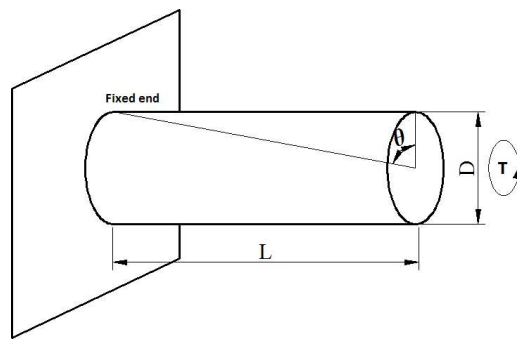


Fig. 2 Shaft under torsion

Procedure:

1. Measure the diameter and gauge length of the specimen as shown in Fig 3.
2. Fix the specimen in the jaws of the machine.
3. Set the pointer on the torque measuring scale to zero.
4. Switch on the machine and start the motor.
5. Note down the torque and angle of twist for every five degrees, till fracture.
6. Plot a graph of torque (T) vs. angle of twist (θ).
7. Now calculate modulus of rigidity, shear stress and torsional stiffness of the specimen.

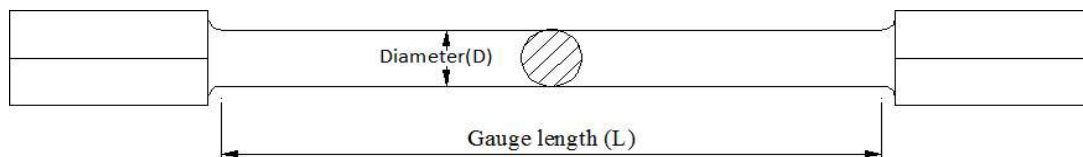


Fig 3 Specimen

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Observation Table No 1:

Sl. No.	Diameter 'D' (mm)				Gauge Length 'L' (mm)	Polar Moment of Inertia 'J' (mm ⁴)
	MSR	VSR	Total	Mean		

Observation Table No 2:

Sl. No.	Angle of Twist 'θ' (rad)	Torque 'T' (N-mm)	Modulus of Rigidity $G = \frac{T}{\theta} \times \frac{L}{J}$ (N/mm ²)	Shear stress $\tau = \frac{G\theta R}{L}$ (N/mm ²)	Torsional Stiffness, $K = \frac{T}{\theta} = \frac{GJ}{L}$ (N-mm/rad)

Conclusion:

From the test, we find that:

Modulus of Rigidity of the specimen = _____ N/mm²

Shear stress of the specimen = _____ N/mm²

Torsional Stiffness of the specimen = _____ N-mm/rad

Ultimate Shear Strength of the specimen = _____ N/mm²

Experiment No. 6

Title: Determination of endurance limit by Fatigue Test

Objective: To determine the endurance limit of the given specimen under fatigue or cyclic loading

Equipments required: Fatigue Testing Machine

Theory:

Fatigue is a form of failure that takes place in structure that are subjected to dynamic and fluctuating stresses, e.g., bridges, aircraft engines and machines. Under these conditions, failure can occur at a stress level considerably lower than the tensile or yield strength. Fatigue failure is just like brittle failure in nature even in normally ductile metals, because there is very little gross plastic deformation associated with failure. Fatigue failure takes place by crack propagation. At some critical stage, the crack propagation becomes rapid culminating in fracture. The fatigue behavior of a material may be understood by analyzing the results of a fatigue test, which are represented in the form of ‘S-N’ curves. The number of cycles of stress reversals ‘N’ required to cause fracture is plotted against the applied stress level ‘S’ in the curve. The fatigue life or endurance limit is the number of cycles to failure at a specified stress level, while the fatigue strength is the stress below which failure does not occur.

The deflection and bending stress that will develop in the specimen during the test is:

Deflection of the specimen, $y = \frac{wL^3}{3EI}$

Bending stress of the specimen, $\sigma = \frac{My}{I}$

where, w = Applied Load

L = Length of the specimen

E = Young’s modulus of the specimen

I = Moment of inertia of the specimen

M = Moment of the specimen

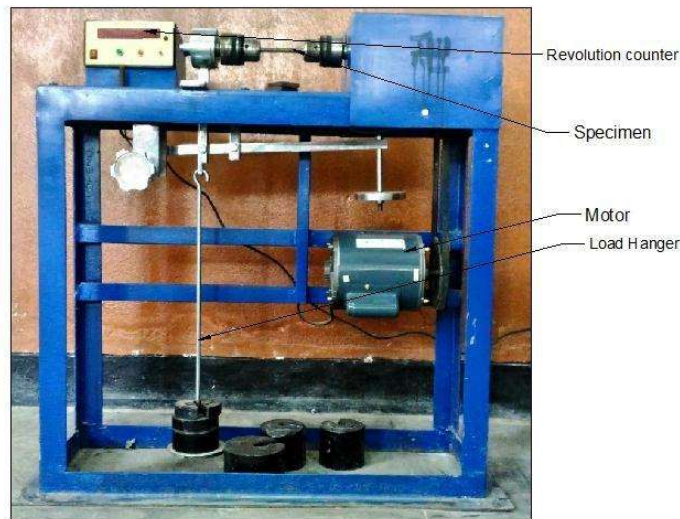


Fig. 1 Fatigue testing machine

Procedure:

1. Measure the gauge length (L) and diameter at the middle portion (d) of the given specimen.
2. Place a suitable load (W) say, 50kg in the weight hanger of the machine and release the cam so as to apply the weight / load to the specimen.
3. Set the electrical counter of the machine to zero position and start the motor of the machine.
4. Record the number indicated on the electrical counter after the specimen failure.
5. Fix another specimen of same material, size and shape in the machine.
6. Repeat the steps from 3 to 8 with 20% decrease in weight each time.
7. Conduct at least 5 tests for a given material.
8. Plot a graph between ' $\log \sigma$ ' vs ' $\log N$ ' by taking $\log \sigma$ along y-axis and $\log N$ along x-axis. The shape of the curve will normally be as shown in Fig. 2.
9. Read the value of stress corresponding to the knee of ' $\sigma - N$ ' curve, which will be the endurance limit or fatigue strength for the given specimen as shown in Fig. 2.

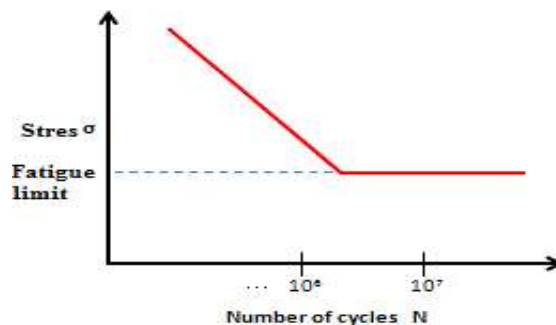


Fig. 2 S-N curve